

Topic 4: Zero-Shot Prompting

4.1 What is Zero-Shot Prompting?

- Zero-shot: asking the model to perform a task without providing any examples
- Relies entirely on the model's pre-trained knowledge and instruction-following ability
- Works well for common tasks the model has seen many times during training
- Fails for highly specialised, unusual, or ambiguous tasks -- use few-shot in those cases
- Zero-shot is the fastest approach -- minimal prompt length, no example overhead

4.2 Effective Zero-Shot Patterns

- Action verb first: 'Classify', 'Summarise', 'Translate', 'Extract', 'List', 'Explain'
- State the task, then the input -- not the other way around
- Specify the audience: 'Explain to a 10-year-old', 'Explain to a senior engineer'
- Specify the length: 'in one sentence', 'in under 100 words', 'in a paragraph'
- Specify the tone: 'formal', 'casual', 'persuasive', 'neutral and factual'

```
WEAK (zero-shot, vague):
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```
What is blockchain?
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```
STRONG (zero-shot, structured):
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```
Explain blockchain technology in 3 sentences.
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```
Audience: a small business owner with no technical background.
```

```
Tone: clear, jargon-free, and practical.
```

4.3 Sentiment and Classification (Zero-Shot)

- LLMs excel at zero-shot classification when categories are clearly named
- Always list the possible categories explicitly
- Ask for just the label to avoid hallucinated reasoning

```
Classify the sentiment of the customer review below.
```

```
Categories: Positive, Negative, Neutral
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```
Return only the category label, nothing else.
```

```
Review: 'The delivery was late but the product quality exceeded my expectations.'
```

```
Expected output: Positive
```

4.4 Limitations of Zero-Shot

- Multi-step reasoning often fails zero-shot -- the model skips steps or makes errors
 - Novel formats the model hasn't seen are unreliable without examples
 - Domain-specific terminology may be misunderstood without context
 - Ambiguous tasks produce inconsistent results across runs
 - Solution: add examples (few-shot) or reasoning steps (chain-of-thought)
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